

## ANSWER KEY & MARKING SCHEME

### Class IX – Sound

**Time:** 60 Minutes

**Maximum Marks:** 40

#### Case Study 1 (1 Mark)

**Answer:** Sound is produced due to **vibrations**.

**Marking:**

- Correct answer – 1

#### Case Study 2 (1 Mark)

**Answer:** There is **no medium (vacuum)** on the Moon to transmit sound.

**Marking:**

- Correct reason – 1

#### Case Study 3 (2 Marks)

- a) Sound **gradually decreases and finally stops** – 1
- b) Sound needs a **material medium** to travel – 1

#### Case Study 4 (2 Marks)

- a) **Frequency** – 1
- b) Unit: **Hertz (Hz)** – 1



**Case Study 5 (2 Marks)**

- a) **Longitudinal wave** – 1  
b) High pressure: **Compression**,  
Low pressure: **Rarefaction** – 1
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**Case Study 6 (2 Marks)**

- a) **Loudness** – 1  
b) SI unit: **decibel (dB)** – 1
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**Case Study 7 (2 Marks)**

- a) **Tuning fork B (512 Hz)** – 1  
b) Higher frequency produces **more shrill sound** – 1
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**Case Study 8 (2 Marks)**

- a) Distance =  $(v \times t)/2 = (340 \times 0.8)/2 = \mathbf{136\ m}$  – 1  
b) Minimum distance should be  $\geq \mathbf{17\ m}$  for distinct echo – 1
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**Case Study 9 (2 Marks)**

- a) Speed =  $f \times \lambda = 400 \times 0.85 = \mathbf{340\ m\ s^{-1}}$  – 1  
b) Medium: **Air** – 1
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**Case Study 10 (2 Marks)**

- a) **Nature / elasticity of the medium – 1**
  - b) Iron has **higher elasticity and particle closeness – 1**
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## Case Study 11 (3 Marks – Tough)

- a) Wavelength =  $v / f = 340 / 500 = 0.68 \text{ m} - 1$
  - b) **Audible sound – 1**
  - c) Used in **music / communication – 1**
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## Case Study 12 (3 Marks – Tough)

- a) Speed of **light is much greater** than sound – 1
  - b) Distance =  $340 \times 5 = 1700 \text{ m} - 1$
  - c) **Longitudinal wave – 1**
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## Case Study 13 (3 Marks – Tough)

- a) **Wave Y (600 Hz) – 1**
  - b) **Both have same energy – 1**
  - c) Energy depends on **amplitude, not frequency – 1**
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## Case Study 14 (3 Marks – HOTS)

- a) **Loudness – 1**
  - b) Amplitude **increases – 1**
  - c) **No, pitch depends on frequency – 1**
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## Case Study 15 (3 Marks – HOTS)



- a) Sound with frequency  $> 20,000 \text{ Hz}$  – 1
  - b) **Imaging internal organs / foetus scanning** – 1
  - c) Ultrasound is **non-ionising and safe** – 1
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## Case Study 16 (3 Marks – HOTS)

- a) **Sound Navigation And Ranging** – 1
  - b) Depth =  $(\text{Speed} \times \text{Time})/2$  – 1
  - c) **Ultrasonic waves** – 1
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## Case Study 17 (3 Marks – HOTS)

- a) Speed **increases** – 1
  - b) Frequency **remains constant** – 1
  - c) Wavelength **increases** – 1
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## Case Study 18 (3 Marks – Conceptual)

- a) Unpleasant and unwanted sound – 1
  - b) Causes **hearing loss / stress / hypertension** – 1
  - c) **Use silencers / limit loudspeakers** – 1
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## Case Study 19 (3 Marks – Numerical)

- a) Frequency =  $680 / 2 = 340 \text{ Hz}$  – 1
  - b) **Audible sound** – 1
  - c) Audible range: **20 Hz – 20,000 Hz** – 1
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## Case Study 20 (3 Marks – Competitive)

- a) Speed =  $1700 / 5 = 340 \text{ m s}^{-1}$  – 1  
 b) Medium: **Air** – 1  
 c) Speed matches **speed of sound in air** – 1

## Overall Mark Distribution

| Level        | Marks     |
|--------------|-----------|
| Easy         | 10        |
| Moderate     | 15        |
| Tough / HOTS | 15        |
| <b>Total</b> | <b>40</b> |

